

STATISTICS REPORT

ACT 25MC1 - SCIENCE



Test Code: 25MC1

School: Sample High School

Grade: 11

Test Date: 10/31/2025

N* = 129

Number of Items = 34

Group: All Students Tested

Strand	Overall	Interpretation of Data (IOD)	Scientific Investigation (SIN)	Evaluation of Models, Inferences, and Experimental Results (EMI)
Score Data				
Number of Items	34	22	5	7
Total Points Possible	34	22	5	7
Maximum Raw Score	30	20	5	7
Minimum Raw Score	13	7	0	1
Range of Raw Scores	17	13	5	6

Assessment Statistics

Mean Raw Score	21.1	14.1	2.8	4.2
Mean Percent Score	62.1%	64.3%	55.2%	59.9%
Met 55% Benchmark Score (% of Students)	72.9%	76.0%	59.7%	70.5%
Standard Deviation	3.41	2.59	1.24	1.53

Item Statistics

ITEM	STRAND CODE	BENCHMARK	STUDENT GROUP	A/F	B/G	C/H	D/J	E/K	BLANK	KEY	Difficulty Index	Discrimination Index
1	IOD Interpretation of Data		Top 27% Students	83%	6%	11%	0%	0%	0%	A	0.57	0.54
	541	Determine and/or use a simple mathematical relationship that exists between data	All Students	57%	22%	14%	7%	0%	0%			
			Bottom 27% Students	29%	40%	23%	9%	0%	0%			
2	IOD Interpretation of Data		Top 27% Students	60%	26%	11%	3%	0%	0%	F	0.55	0.11
	451	Compare or combine data from a simple data presentation (e.g., order or sum data from a table)	All Students	55%	22%	15%	8%	0%	0%			
			Bottom 27% Students	49%	17%	23%	11%	0%	0%			
3	IOD Interpretation of Data		Top 27% Students	3%	11%	6%	77%	0%	3%	D	0.63	0.23
	221	Identify basic features of a table, graph, or diagram in a simple presentation	All Students	5%	15%	9%	63%	8%	2%			
			Bottom 27% Students	3%	14%	11%	54%	14%	3%			

Items 29, 30, 31, 32, 33 and 34 are field-tested and not included in the scoring.

***Use the data with caution when N < 10.**

ITEM	STRAND CODE	BENCHMARK	STUDENT GROUP	A/F	B/G	C/H	D/J	E/K	BLANK	KEY	Difficulty Index	Discrimination Index
4	IOD Interpretation of Data		Top 27% Students	3%	6%	80%	11%	0%	0%	H	0.65	0.23
	541	Determine and/or use a simple mathematical relationship that exists between data	All Students	7%	17%	65%	10%	0%	1%			
			Bottom 27% Students	9%	26%	57%	9%	0%	0%			
5	IOD Interpretation of Data		Top 27% Students	3%	9%	6%	71%	9%	3%	D	0.63	0.14
	321	Understand basic scientific terminology	All Students	5%	15%	9%	63%	8%	2%			
			Bottom 27% Students	6%	20%	9%	57%	9%	0%			
6	IOD Interpretation of Data		Top 27% Students	86%	6%	9%	0%	0%	0%	F	0.57	0.60
	421	Translate information into a table, graph, or diagram	All Students	57%	22%	14%	7%	0%	0%			
			Bottom 27% Students	26%	40%	26%	9%	0%	0%			
7	EMI Evaluation of Models, Inferences, and Experimental Results		Top 27% Students	11%	9%	80%	0%	0%	0%	C	0.58	0.34
	311	Determine which models present certain basic information	All Students	26%	12%	58%	4%	0%	0%			
			Bottom 27% Students	37%	9%	46%	9%	0%	0%			
8	IOD Interpretation of Data		Top 27% Students	3%	6%	6%	80%	6%	0%	J	0.60	0.40
	221	Identify basic features of a table, graph, or diagram in a simple presentation	All Students	8%	14%	9%	60%	9%	1%			
			Bottom 27% Students	6%	31%	11%	40%	11%	0%			
9	EMI Evaluation of Models, Inferences, and Experimental Results		Top 27% Students	0%	6%	83%	11%	0%	0%	C	0.61	0.51
	311	Determine which models present certain basic information	All Students	9%	19%	61%	11%	0%	1%			
			Bottom 27% Students	6%	34%	31%	26%	0%	3%			
10	EMI Evaluation of Models, Inferences, and Experimental Results		Top 27% Students	66%	14%	11%	9%	0%	0%	F	0.57	0.14
	311	Determine which models present certain basic information	All Students	57%	22%	13%	8%	0%	0%			
			Bottom 27% Students	51%	26%	14%	9%	0%	0%			
11	EMI Evaluation of Models, Inferences, and Experimental Results		Top 27% Students	0%	6%	83%	11%	0%	0%	C	0.61	0.51
	421	Identify similarities and differences between models	All Students	9%	18%	61%	11%	0%	1%			
			Bottom 27% Students	6%	34%	31%	26%	0%	3%			
12	IOD Interpretation of Data		Top 27% Students	3%	77%	11%	9%	0%	0%	G	0.74	0.17
	341	Determine how the values of dependent variables change in a simple data presentation	All Students	6%	74%	14%	6%	0%	0%			
			Bottom 27% Students	11%	60%	20%	9%	0%	0%			

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ITEM	STRAND CODE	BENCHMARK	STUDENT GROUP	A/F	B/G	C/H	D/J	E/K	BLANK	KEY	Difficulty Index	Discrimination Index
13	EMI Evaluation of Models, Inferences, and Experimental Results		Top 27% Students	26%	9%	66%	0%	0%	0%	C	0.63	0.06
	431	Determine which simple hypothesis is, or is not, consistent with data or model	All Students	24%	11%	63%	2%	0%	0%			
			Bottom 27% Students	31%	9%	60%	0%	0%	0%			
14	SIN Scientific Investigation		Top 27% Students	51%	26%	14%	9%	0%	0%	H	0.29	-0.11
	541	Determine the experimental conditions that would produce specified results	All Students	38%	19%	29%	12%	0%	2%			
			Bottom 27% Students	43%	14%	26%	17%	0%	0%			
15	IOD Interpretation of Data		Top 27% Students	3%	77%	14%	6%	0%	0%	B	0.72	0.11
	541	Determine and/or use a simple mathematical relationship that exists between data	All Students	5%	72%	14%	9%	0%	0%			
			Bottom 27% Students	9%	66%	17%	9%	0%	0%			
16	IOD Interpretation of Data		Top 27% Students	14%	0%	66%	20%	0%	0%	H	0.61	0.11
	541	Determine and/or use a simple mathematical relationship that exists between data	All Students	9%	18%	61%	12%	0%	1%			
			Bottom 27% Students	6%	31%	54%	9%	0%	0%			
17	SIN Scientific Investigation		Top 27% Students	3%	74%	17%	6%	0%	0%	B	0.69	0.14
	331	Find basic information in text that describes a complex experiment	All Students	6%	69%	16%	9%	0%	0%			
			Bottom 27% Students	6%	60%	20%	14%	0%	0%			
18	SIN Scientific Investigation		Top 27% Students	77%	6%	14%	3%	0%	0%	F	0.57	0.26
	321	Understand the methods used in a simple experiment	All Students	57%	22%	13%	8%	0%	0%			
			Bottom 27% Students	51%	26%	17%	6%	0%	0%			
19	IOD Interpretation of Data		Top 27% Students	6%	9%	11%	63%	11%	0%	D	0.60	0.06
	441	Perform a simple interpolation or simple extrapolation using data in a table or graph	All Students	5%	14%	9%	60%	9%	2%			
			Bottom 27% Students	9%	11%	11%	57%	9%	3%			
20	EMI Evaluation of Models, Inferences, and Experimental Results		Top 27% Students	83%	9%	9%	0%	0%	0%	F	0.57	0.54
	311	Determine which models present certain basic information	All Students	57%	22%	14%	7%	0%	0%			
			Bottom 27% Students	29%	40%	23%	9%	0%	0%			
21	EMI Evaluation of Models, Inferences, and Experimental Results		Top 27% Students	14%	17%	69%	0%	0%	0%	C	0.63	0.26
	311	Determine which models present certain basic information	All Students	23%	11%	63%	3%	0%	0%			
			Bottom 27% Students	40%	9%	43%	9%	0%	0%			

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22	IOD Interpretation of Data		Top 27% Students	6%	11%	6%	69%	9%	0%	J	0.62	0.11
	221	Identify basic features of a table, graph, or diagram in a simple presentation	All Students	6%	15%	7%	62%	9%	1%			
			Bottom 27% Students	6%	14%	6%	57%	17%	0%			
23	SIN Scientific Investigation		Top 27% Students	9%	6%	86%	0%	0%	0%	C	0.63	0.34
	311	Understand the tools and functions of tools used in a complex experiment	All Students	24%	11%	63%	2%	0%	0%			
			Bottom 27% Students	34%	9%	51%	6%	0%	0%			
24	IOD Interpretation of Data		Top 27% Students	0%	9%	20%	63%	9%	0%	J	0.60	0.17
	231	Find basic information in text that describes a simple data presentation	All Students	8%	14%	10%	60%	9%	0%			
			Bottom 27% Students	14%	17%	9%	46%	14%	0%			
25	IOD Interpretation of Data		Top 27% Students	3%	77%	17%	3%	0%	0%	B	0.72	0.14
	441	Perform a simple interpolation or simple extrapolation using data in a table or graph	All Students	5%	72%	15%	8%	0%	0%			
			Bottom 27% Students	11%	63%	9%	17%	0%	0%			
26	IOD Interpretation of Data		Top 27% Students	3%	14%	0%	77%	6%	0%	J	0.62	0.26
	551	Compare or combine data from two or more simple data presentations	All Students	5%	15%	8%	62%	9%	1%			
			Bottom 27% Students	6%	17%	6%	51%	17%	3%			
27	IOD Interpretation of Data		Top 27% Students	3%	6%	3%	83%	6%	0%	D	0.61	0.43
	541	Determine and/or use a simple mathematical relationship that exists between data	All Students	5%	15%	8%	61%	9%	2%			
			Bottom 27% Students	6%	34%	9%	40%	11%	0%			
28	IOD Interpretation of Data		Top 27% Students	6%	71%	14%	9%	0%	0%	G	0.70	0.03
	541	Determine and/or use a simple mathematical relationship that exists between data	All Students	6%	70%	14%	10%	0%	0%			
			Bottom 27% Students	6%	69%	20%	6%	0%	0%			
35	IOD Interpretation of Data		Top 27% Students	3%	86%	9%	3%	0%	0%	B	0.72	0.34
	551	Compare or combine data from two or more simple data presentations	All Students	6%	72%	14%	8%	0%	0%			
			Bottom 27% Students	11%	51%	23%	14%	0%	0%			
36	SIN Scientific Investigation		Top 27% Students	80%	6%	14%	0%	0%	0%	F	0.58	0.31
	321	Understand the methods used in a simple experiment	All Students	58%	21%	14%	7%	0%	0%			
			Bottom 27% Students	49%	26%	20%	6%	0%	0%			

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37	IOD Interpretation of Data		Top 27% Students	6%	77%	11%	6%	0%	0%	B	0.74	0.14
	231	Find basic information in text that describes a simple data presentation	All Students	6%	74%	15%	5%	0%	0%			
			Bottom 27% Students	6%	63%	20%	11%	0%	0%			
38	IOD Interpretation of Data		Top 27% Students	3%	86%	9%	3%	0%	0%	G	0.71	0.34
	541	Determine and/or use a simple mathematical relationship that exists between data	All Students	6%	71%	15%	8%	0%	0%			
			Bottom 27% Students	11%	51%	26%	11%	0%	0%			
39	IOD Interpretation of Data		Top 27% Students	6%	9%	80%	6%	0%	0%	C	0.61	0.23
	421	Translate information into a table, graph, or diagram	All Students	10%	17%	61%	11%	0%	1%			
			Bottom 27% Students	6%	23%	57%	14%	0%	0%			
40	IOD Interpretation of Data		Top 27% Students	3%	11%	6%	71%	9%	0%	J	0.62	0.17
	541	Determine and/or use a simple mathematical relationship that exists between data	All Students	4%	15%	9%	62%	10%	1%			
			Bottom 27% Students	11%	17%	11%	54%	3%	3%			

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